Scale-Consistent Mean Curvature of Osteon Formation-Front Contours

OPALSx — osteon profile and level-set analysis

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Abstract

The per-osteocyte curvature pipeline measures the curvature of a 2-D formation-front contour over a fixed *physical* arc length k_{scale} (`k_scale_um`). We observed that the reported *contour-mean* curvature varies strongly with k_{scale} for late-forming (small, inner) contours, while being stable for early (large) ones. This note shows that the cause is a systematic bias of the local parabola-fitting estimator whose relative size grows as $(k_{\text{scale}}/L)^2$, where L is the contour perimeter. The remedy is to treat the contour mean as a *global* quantity rather than averaging scale-dependent local estimates. We give two window-free estimators that are exactly independent of k_{scale} and agree with the small-window limit of the local estimate: the *turning fit* $\overline{\kappa} = 2\pi/L$ (from the total turning, Hopf's *Umlaufsatz*) and the *circle fit* $\overline{\kappa} = 1/R$ (a circle fitted about the contour centroid). Both are implemented behind a single `method` choice (`:ALF`/`:CCF`/`:CTF`, default `:CCF`).

1 Setup and notation

At an osteocyte's formation time t the level-set field $\phi(t) = (1-t)\, S_{\text{out}} - t\, S_{\text{in}}$ is built (and Gaussian smoothed) on the osteocyte's slice, and its zero contour $\mathcal{C} = \{(x, y) : \phi = 0\}$ is extracted as a closed polygon with vertices $P_1, \dots, P_N$ (with $P_{N+1} = P_1$), in physical units (µm).

For a smooth plane curve the *signed curvature* is

$$\kappa \;=\; \frac{x'\, y'' - y'\, x''}{\left(x'^2 + y'^2\right)^{3/2}}, \tag{1}$$

and, for a graph $y = f(x)$,

$$\kappa \;=\; \frac{f''}{\left(1 + f'^2\right)^{3/2}}. \tag{2}$$

Local estimator (`compute_2D_curvature`). At each vertex P_i the contour is translated to the origin and rotated so the local tangent lies along the x-axis. A least-squares parabola $y = ax^2 + bx + c$ is fitted to the vertices lying within an arc-length window of total length k_{scale} (i.e. $\pm h$, $h = k_{\text{scale}}/2$, on each side). At the central point $x = 0$ one has $f'(0) = b$, $f''(0) = 2a$, so by (2)

$$\widehat{\kappa}_i \;=\; \frac{2a}{\left(1 + b^2\right)^{3/2}}. \tag{3}$$

The window k_{scale} fixes the *physical scale* at which curvature is measured, so the same scale is used on contours of very different size.

Contour mean (as previously computed). The diagnostics reported the contour mean as the unweighted vertex average

$$\langle \kappa \rangle = \frac{1}{N} \sum_{i=1}^{N} \widehat{\kappa}_i. \qquad (4)$$

2 The observed problem

Sweeping k_{scale} over $[10, 150]$ µm on a single dataset (`FM40-2-S2`) gives an early/large contour whose mean is nearly constant, but a late/small contour whose mean inflates by more than a factor of two before collapsing (Table 1, Figure 1).

idx	t	L [µm]	$2\pi/L$ [µm^{-1}]	$\langle\kappa\rangle$ mean [µm^{-1}]	CV over k
3	0.05	1009.9	0.00622	0.00631	2.8 %
20	0.32	733.9	0.00856	0.00889	2.6 %
37	0.92	179.3	0.03503	0.05436	28.3 %

Table 1: Mean curvature of three contours, swept over $k_{\text{scale}} \in [10, 150]$ µm in steps of 10 µm. The coefficient of variation (CV = std/mean across the sweep) is small for the large contours but explodes for the small late contour. $2\pi/L$ is the scale-invariant value derived in Section 4.

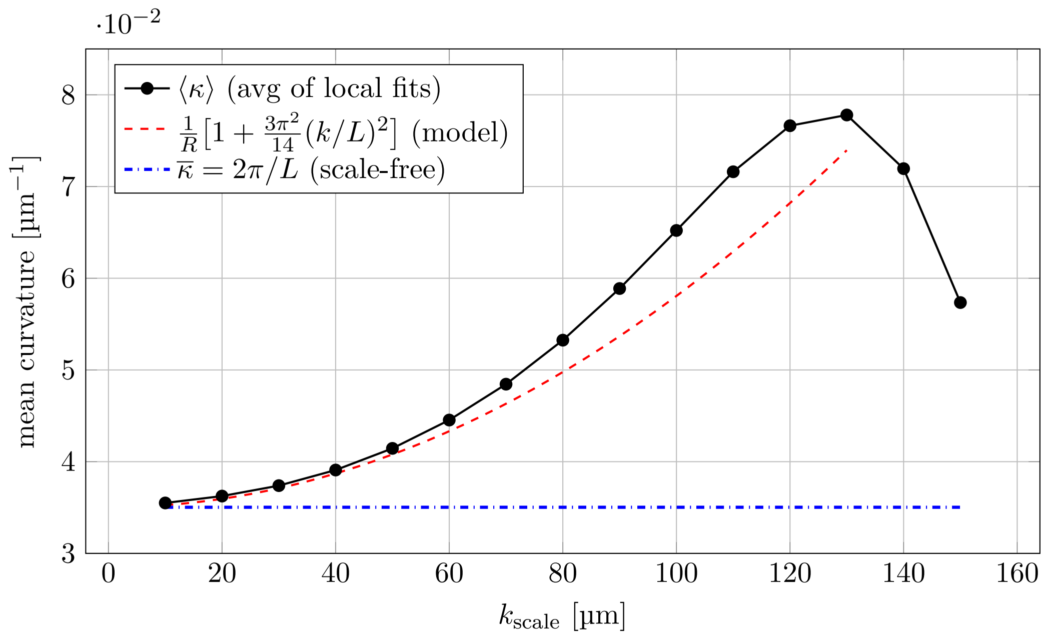

Figure 1: Late contour (idx 37, $L = 179.3$ µm). The averaged local estimate (black) departs from the true mean $2\pi/L$ (blue) as the window grows, tracking the leading-order $(k/L)^2$ bias (red) until the window becomes a large fraction of the perimeter and the fit degenerates entirely.

Two facts narrow down the cause:

1. At small k_{scale} the mean converges to $2\pi/L$ ($0.0355 \to 0.0350$ for idx 37); the inflation appears only as the window grows.

2. Replacing the unweighted average (4) with the arc-length-weighted average $\sum_i \widehat{\kappa}_i \, \Delta s_i / \sum_i \Delta s_i$ changes nothing (CV stays 28 %): marching-squares vertices are near-uniformly spaced, so the two averages coincide. The problem is therefore in the per-vertex estimates $\widehat{\kappa}_i$ themselves, not in how they are averaged.

3 Why the local estimator is scale-dependent

Consider the cleanest case: $\mathcal{C}$ is a circle of radius R, so the true curvature is $1/R$ everywhere. In the tangent frame at the fitting point the circle is the graph

$$f(x) \;=\; R - \sqrt{R^2 - x^2} \;=\; \frac{x^2}{2R} + \frac{x^4}{8R^3} + \frac{x^6}{16R^5} + \cdots \tag{5}$$

The estimator (3) fits only a quadratic, so the quartic and higher terms in (5) leak into the fitted coefficient a and bias it. By symmetry $b = 0$, and the least-squares estimate of a is obtained by projecting f onto $\{1, x^2\}$ over the fitting interval $[-X, X]$, where $X = R \sin \Phi$ is the half-width in x and Φ the half-arc subtended at the centre.

Regressing x^4 onto $\{1, x^2\}$ over $[-X, X]$ (uniform measure) gives the x^2-coefficient

$$q \;=\; \frac{M_0 M_6 - M_2 M_4}{M_0 M_4 - M_2^2} \;=\; \frac{\frac{4}{7} - \frac{4}{15}}{\frac{4}{5} - \frac{4}{9}}\, X^2 \;=\; \frac{6}{7}\, X^2, \qquad M_{2k} = \int_{-X}^{X} x^{2k}\, dx, \tag{6}$$

so to leading order

$$\widehat{a} \;=\; \frac{1}{2R} + \frac{1}{8R^3} \cdot \frac{6}{7} X^2 + \cdots, \qquad \widehat{\kappa} \;=\; 2\widehat{a} \;=\; \frac{1}{R}\left[1 + \frac{3}{14}\Big(\frac{X}{R}\Big)^2 + \cdots\right]. \tag{7}$$

The bias is positive: a parabola fitted over a wide arc must reach the far, “flattening” points of the circle, which it can only do by curving more sharply, so $\widehat{\kappa} > 1/R$. (In the extreme $\Phi \to \pi/2$, the far point is (R, R) and a parabola through it has $aR^2 = R$, i.e. $\widehat{\kappa} = 2/R$, double the truth.)

Linking the window to the contour size. The half-window in arc length is $h = k_{\text{scale}}/2 = R\Phi$, so $\Phi = k_{\text{scale}}/(2R)$. Writing the radius in terms of the perimeter via the effective radius $R_{\text{eff}} = L/2\pi$ (exact for a circle, and the natural size scale for a general convex contour, Section 4),

$$\Phi \;=\; \frac{k_{\text{scale}}}{2R_{\text{eff}}} \;=\; \pi\, \frac{k_{\text{scale}}}{L}. \tag{8}$$

For modest windows $X/R \approx \Phi$, and (7) becomes a clean law in the dimensionless ratio k_{scale}/L:

$$\boxed{\frac{\widehat{\kappa}}{\kappa_{\text{true}}} \;\approx\; 1 + \frac{3}{14}\,\Phi^2 \;=\; 1 + \frac{3\pi^2}{14}\Big(\frac{k_{\text{scale}}}{L}\Big)^2 \;\approx\; 1 + 2.11\Big(\frac{k_{\text{scale}}}{L}\Big)^2.} \tag{9}$$

This explains every feature of the data:

- the bias is *quadratic* in k_{scale}/L, hence negligible for large early contours ($k_{\text{scale}}/L \ll 1$) and large for small late contours;
- for idx 37 at $k_{\text{scale}} = 60\,\mu\text{m}$, $k_{\text{scale}}/L = 0.33$ and (9) predicts +24 % (observed +27 %; the remaining excess is the neglected x^6 and higher terms, which steepen the curve);
- once $k_{\text{scale}} \sim L$ the window wraps most of the contour, the fit is meaningless, and the curve in Figure 1 turns over.

Crucially, the bias (9) is a property of *each* $\widehat{\kappa}_i$. No reweighting of the $\widehat{\kappa}_i$ can remove it — which is exactly why the arc-length-weighted average behaves identically to the unweighted one.

4 Two scale-invariant alternatives

The mean curvature of a *closed* contour need not be estimated locally at all. We give two window-free estimators; both reduce to $1/R$ for a circle and neither depends on k_{scale}.

4.1 Contour turning fit (`:CTF`)

For a positively oriented, simple, closed C^2 curve, the theorem of turning tangents (Hopf's *Umlaufsatz*) states

$$\oint_{\mathcal{C}} \kappa \, ds \;=\; 2\pi \, n, \tag{10}$$

where n is the turning (rotation) number; $n = 1$ for a simple convex contour. The *arc-length-weighted* mean curvature is therefore independent of any measurement window:

$$\boxed{\overline{\kappa}_{\text{CTF}} \;=\; \frac{\oint_{\mathcal{C}} \kappa \, ds}{\oint_{\mathcal{C}} ds} \;=\; \frac{2\pi n}{L} \;=\; \frac{2\pi}{L} \;=\; \frac{1}{R_{\text{eff}}} \quad \text{(simple contour).}} \tag{11}$$

This is the value $\langle \kappa \rangle$ converges to as $k_{\text{scale}} \to 0$ in Figure 1: from (9), the arc-length integral of the *unbiased* curvature is 2π, while at finite window the integral of the inflated $\widehat{\kappa}$ exceeds 2π, pushing the average above $2\pi/L$. Discretely, for the polygon $P_1, \ldots, P_N$ with edge vectors $\mathbf{e}_i = P_{i+1} - P_i$, the integral (10) is a sum of exterior (turning) angles,

$$\theta_i \;=\; \operatorname{atan2}\!\big(\mathbf{e}_{i-1} \times \mathbf{e}_i,\; \mathbf{e}_{i-1} \cdot \mathbf{e}_i\big), \qquad \sum_{i=1}^{N} \theta_i \;=\; 2\pi n, \qquad \overline{\kappa}_{\text{CTF}} \;=\; \frac{\sum_i \theta_i}{\sum_i |\mathbf{e}_i|}, \tag{12}$$

which holds exactly for any simple closed polygon regardless of vertex spacing (implemented as `turning_fit_curvature`).

4.2 Contour circle fit (`:CCF`, default)

Alternatively, fit a circle to all contour vertices with its centre fixed at the centroid $\bar{P} = \frac{1}{N} \sum_i P_i$, and read the mean curvature as the inverse of the fitted radius. With the centre fixed, the least-squares radius minimises

$$E(R) \;=\; \sum_{i=1}^{N} \big(r_i - R\big)^2, \qquad r_i = \|P_i - \bar{P}\|, \tag{13}$$

and, since $E'(R) = -2 \sum_i (r_i - R) = 0$, has the closed form "mean radial distance":

$$\boxed{R \;=\; \frac{1}{N} \sum_{i=1}^{N} \|P_i - \bar{P}\|, \qquad \overline{\kappa}_{\text{CCF}} \;=\; \frac{1}{R}.} \tag{14}$$

For a circle the P_i are equidistant from the centre $= \bar{P}$, so R is the true radius and $\overline{\kappa}_{\text{CCF}} = 1/R$ exactly. Like the turning fit, (14) uses no measurement window (implemented as `circle_fit_curvature`).

How the two differ. Both write $\overline{\kappa} = 1/R_{\text{eff}}$ but use different effective radii,

$$R_{\text{eff}}^{\text{CTF}} = \frac{L}{2\pi} \quad \text{(perimeter radius)}, \qquad R_{\text{eff}}^{\text{CCF}} = \frac{1}{N}\sum_i \|P_i - \bar{P}\| \quad \text{(radial-mean radius)}. \tag{15}$$

They coincide for a circle. For a wiggly boundary the perimeter radius is sensitive to small-scale roughness (crenellation inflates L and hence $\overline{\kappa}_{\text{CTF}}$), whereas the radial-mean radius averages such roughness out but presumes the contour is reasonably centred and round. For the near-circular osteon contours the two agree closely (Section 5).

5 Verification

Both window-free estimators reproduce $1/R$ and agree with the reliable small-window limit of the local estimator, with zero scale dependence:

idx	t	turning $\frac{2\pi}{L}$	circle $\frac{1}{R}$	small-k (k=10)	averaged $\langle\kappa\rangle$ CV
3	0.05	0.00622	0.00649	0.00646	2.8 %
20	0.32	0.00856	0.00884	0.00863	2.6 %
37	0.92	0.03503	0.03510	0.03550	28.3 %

The turning ($2\pi/L$) and circle ($1/R$) means agree with each other and with the (reliable) small-window estimate to within a few percent on every contour, while the windowed average drifts by up to 28 % on the small late contour. On a perfect circle of radius 30 µm all three methods return $1/R = 0.0333\,\mu\text{m}^{-1}$ (the average of local fits at $k_{\text{scale}} = 20\,\mu\text{m}$ gives 0.0342, the expected $\sim$2 % window bias).

6 Implementation and recommendations

The three estimators are exposed through a single method choice,

```
contour_mean_curvature(x, y; method=:CCF, arclen=nothing),
```

with `method` one of:

- `:ALF` — *average of local fits*, $\langle\kappa\rangle$ (4); scale-dependent, uses the window `arclen` (= k_{scale});
- `:CCF` — *contour circle fit* (14); scale-free **(default)**;
- `:CTF` — *contour turning fit* (12); scale-free.

The same `mean_method` keyword is threaded through `compute_curvature_near_osteocyte` (which returns `mean_available_κ`) and `plot_osteocyte_contour`; both default to `:CCF`.

Recommendations.

- **Report the contour mean with `:CCF` (default) or `:CTF`.** Both are exact for a circle, robust to sampling, and free of the $(k_{\text{scale}}/L)^2$ bias of `:ALF`.
- **Keep the windowed per-vertex estimator (`compute_2D_curvature`) for *local* curvature** (e.g. at the osteocyte), where a fixed physical scale is the intent. To hold its bias below $\sim$2 %, choose $k_{\text{scale}}/L \lesssim 0.1$ for the *smallest* contour of interest (from (9), $2.11 \times 0.1^2 \approx 0.02$).

- **Interpretational caveat.** Both window-free means are *size* measures ($1/R_{\text{eff}}$): by (10) every simple closed contour has total turning 2π regardless of local concavities, so neither $\overline{\kappa}_{\text{CTF}}$ nor $\overline{\kappa}_{\text{CCF}}$ can distinguish convex- from concave-preferring embedding. That question is inherently *local* and must be read from the per-vertex $\widehat{\kappa}_i$ (e.g. κ at the osteocyte relative to $\overline{\kappa}$), measured at an appropriately small k_{scale}.